

CELLGUARD · DESIGN PRECIS

CellGuard design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellguard		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Key numbers from CGD-CAL-001 v0.6 (1.5 mOhm-class switches, five-day state-of-charge error); status light and light pipe in the components table; cost USD 159.15 against the USD 140 value-engineering target
0.7	2026-10-02	Amish Chadha	Responses to R6, R8 and R14, cover, status light, state-of-charge check and first test partner as decided on 2026-10-02 (CGD-DEC-001)
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.5	2026-09-30	Amish Chadha	Design for construction (CGD-DDR-003); figures follow CGD-CAL-001 v0.3; base plate, fuse holder and cover fixings made buildable; GA Rev P2
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. Record design choices adopted for TRL 3 under Amish's 2026-09-25

instruction, open for his review (CGD-DDR-001); add the secondary protector; numbers replaced by CGD-CAL-001; media from the parametric model

0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

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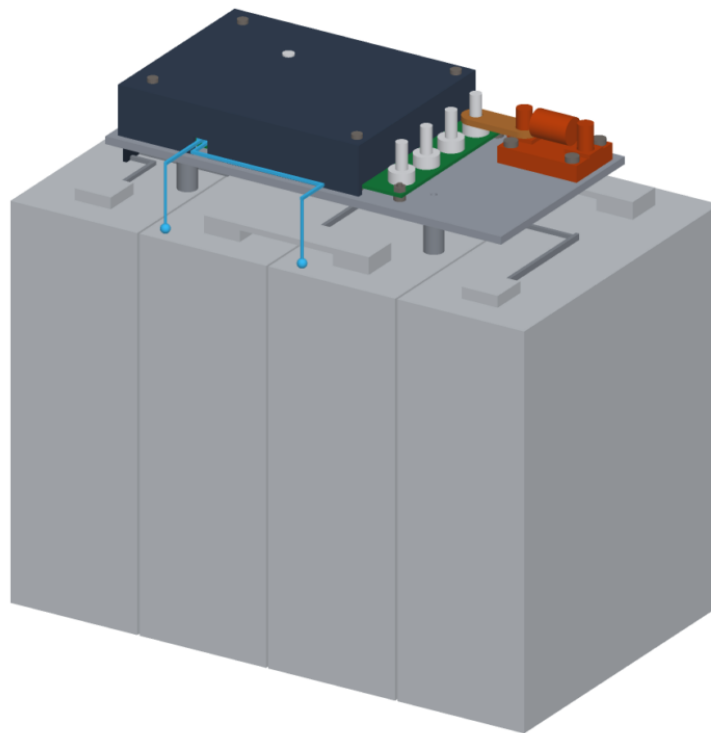
■ Summary

CellGuard is one open battery management board for LFP packs of 4 to 16 cells in series. A single cell-monitor and protection chip measures every cell, trips the pack's MOSFET switches on its own when a limit is crossed, and balances the cells; a small microcontroller adds the state-of-charge estimate, a state-of-health log and an open CAN and UART interface. An independent secondary protector blows a self-control protector (SCP) fuse if the main protection fails. Everything sits on an aluminium base plate that spreads heat and also carries the pack fuse, so the whole protection path is one part that other lab projects can bolt on. The TRL 3 calculations (CGD-CAL-001 v0.6) show 12 of 16 requirements met on paper after the responses Amish decided on 2026-10-02: board loss 3.5 W against 4 W with 1.5 mΩ-class switches (R6), state of charge within 10 points between full charges prompted about every five days (R8, with calibration) and mass 0.66 kg against the relaxed 0.7 kg (R14). R2, R4 and R7 are at risk. Value-engineering target: USD 140. Estimated cost of the constructable design: USD 159.15 (USD 19.15 over the target), mainly the dearer switches (R16).

CellGuard

github.com/BoujeeEnjinia1701/cellguard

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: 4S LiFePO4 pack (4 x 280 Ah cells), mounting posts and cell tap leads for scale

Figure 1. CellGuard on four posts above a 4S LFP pack of 280 Ah prismatic cells, shown for scale. Generated from `cad/src/model.py`. Concept, not for fabrication.

■ How it works

1. **Measure.** The front end (BOM 3) reads every cell voltage through the balance harness (10), pack current through a 0.25 mΩ shunt (5) and three temperatures (11): two cell probes and one thermistor beside the MOSFETs.
2. **Protect.** The front end compares each reading with thresholds loaded from the configuration file at startup. If a cell goes above 3.65 V it opens the charge MOSFETs; below 2.50 V it opens the discharge MOSFETs; on overcurrent, short circuit or temperature faults it opens both (4). This happens in the front end's own hardware, so a firmware crash in the microcontroller does not stop protection. A separate secondary protector (14) watches every cell as well and, if the front end or a MOSFET fails, blows the SCP fuse in the B+ path.
3. **Balance.** When the pack is near full and cells differ by more than a set amount (default 10 mV), the front end bleeds up to 100 mA from the higher cells through external balance resistors.
4. **Connect safely.** The discharge path stays off until the host pulls the enable line. The precharge switch and resistor (7) first charge the host's input capacitors, then the main discharge MOSFETs close. The pack fuse (8) is the last line of defense against a short circuit that the MOSFETs cannot clear.
5. **Estimate and report.** The microcontroller (6) counts charge in and out, corrects the count at full charge and near empty, logs every charge and discharge event to flash and publishes status on CAN and UART (12).

CellGuard: power path, 16S pack discharging at 40 A (estimates, CGD-CAL-001)

github.com/BoujeeEnjinia1701/cellguard

CONCEPT, NOT FOR FABRICATION

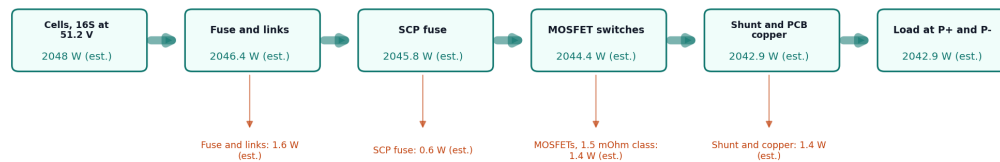


Figure 2. Power path for a 16S pack discharging at 40 A. All values are estimates from CGD-CAL-001.

■ Main components

Table 1. Main components. Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

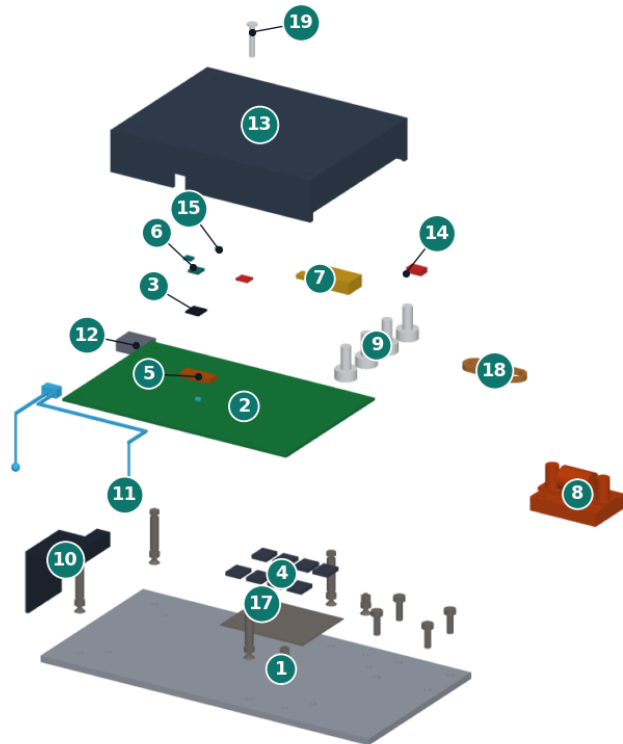
BOM	Component	Concept choice	Notes
1	Base plate and heat spreader	6061 aluminium, 220 x 110 x 4 mm, flat, drilled and tapped; the board sits on six bought 3 mm spacers (CGD-DDR-003)	Takes MOSFET heat through a gap pad; carries the fuse
2	Main PCB	4-layer, 2 oz (70 µm) copper, 150 x 95 mm	Power copper on inner layers; clearances for 60 V
3	Cell monitor and protection IC	TI BQ76952 : 3 to 16 cells, ±15 mV total cell voltage error from –40 to 85 °C, integrated high-side N-channel MOSFET driver, coulomb counter, I ² C	Decided by Amish, 2026-09-25 (CGD-DDR-001 item 1, CGD-DDR-002)
4	Charge and discharge MOSFETs	8 x 100 V N-channel, 1.5 mΩ class (decided 2026-10-02, CGD-DEC-001), top-side cooled (TOLT) package (decided, CGD-DDR-002): 4 in parallel for charge, 4 for discharge, back to back	Mounted on the PCB underside, pressed onto the base plate through an insulating pad
5	Current shunt	0.25 mΩ metal element, 3 W, low side between B- and P-	10 mV at 40 A
6	Microcontroller and CAN transceiver	STM32G0B1 class with built-in CAN controller, plus a 3.3 V CAN transceiver	Runs state of charge, log and interface; not in the protection path. Decided by Amish, 2026-09-25 (item 2)
7	Precharge	100 Ω 10 W aluminium-housed resistor with a DPAK-class P-channel MOSFET, driven by the front end's precharge output	1 s timeout in firmware; DPAK-class switch sized for a 34 W peak (decided, CGD-DDR-002)
8	Pack fuse and holder	60 A, DC rated 80 V or more, breaking capacity 10 kA or more	In the B+ lead, bolted to the base plate
9	Power terminals	Four M6 studs: B+, B-, P+, P-	Pack side (B) and load or charger side (P)
10	Balance connector and harness	17-way connector, 16 sense leads plus B- reference, each lead fused or resistor-protected at the cell end	Unused inputs linked on the board for packs under 16S
11	Temperature sensors	Three 10 kΩ NTC thermistors: two ring-lug probes on cells, one on the board by the MOSFETs	Charge locked out below 0 °C
12	CAN and UART connector	Latching 6-way connector: CAN-H, CAN-L, UART TX and RX, enable, ground	Enable keeps the output off until the host is connected
13	Cover	Flame-retardant polycarbonate (UL 94 V-0 grade), printed, opaque (decided 2026-10-02)	Leaves the studs and fuse reachable; a light pipe (line 19) shows the status light
14	Secondary protector and SCP fuse	TI BQ77216 class: 3 to 16 cells, overvoltage, undervoltage, open wire and temperature, separate COUT and DOUT outputs, about 1 µA; drives a self-control protector fuse in the B+ path	Decided by Amish, 2026-09-25 (item 3); SCP rating at 40 A and 60 V to be confirmed
15	Supporting electronics	60 V input buck regulator, 16 Mbit SPI flash, TVS diode across P+ and P-, balance resistors, passives, and one three-colour status light (decided 2026-10-02)	Only the status light is shown in the model

BOM	Component	Concept choice	Notes
19	Light pipe	Round flanged light pipe, 3 mm, pressed into the cover top over the status light (decided 2026-10-02)	Lets the opaque cover show the status light

CellGuard: exploded viewgithub.com/BoujeeEnjinia1701/cellguard

CONCEPT, NOT FOR FABRICATION

- 1 Base plate and heat spreader
- 2 Main PCB, 4-layer, 2 oz copper
- 3 Cell monitor and protection IC
- 4 Charge and discharge MOSFETs (8)
- 5 Current shunt, 0.25 mOhm
- 6 Microcontroller and CAN transceiver
- 7 Precharge resistor and switch
- 8 Pack fuse and holder
- 9 Power terminals (B-, P-, P+, B+)
- 10 Balance connector and lead harness
- 11 Temperature sensors (3)
- 12 CAN and UART connector
- 13 Flame-retardant cover
- 14 Secondary protector
- 15 Status light, three-colour
- 17 Gap pad, spacers, pillars and screws
- 18 Fuse link (copper bar)
- 19 Light pipe



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 3. Exploded view with BOM numbers.

CellGuard: cutawaygithub.com/BoujeeEnjinia1701/cellguard

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 4. Section through the board: MOSFETs (4) under the PCB on the base plate (1), precharge resistor (7), studs (9) and fuse (8), all under or beside the cover (13).

The general arrangement is drawing CGD-DWG-001 Rev P2 ([cad/drawings/CGD-DWG-001.pdf](#)), and the STEP files are in [cad/step/](#).

■ Key design choices

The choices below were recommended at TRL 2 and are decided by Amish, 2026-09-25: go with recommendation (CGD-DDR-001 as updated, CGD-DDR-002). This includes the licensing rule in choice 3 and the protection settings, top-side cooled MOSFETs, DPAK-class precharge switch and one-point current calibration from CGD-CAL-001.

1. **One analog front end for 4 to 16 cells.** A single BQ76952-class chip covers the whole cell range on one board and runs protection without the microcontroller. Alternatives: an Analog Devices ADBMS or LTC681x daisy-chain monitor (better for large packs, but needs a separate protection path and more firmware), or a Renesas ISL94202 as on the Libre Solar board (autonomous, but limited to 8 cells). Decided: BQ76952 class.
2. **High-side switching.** N-channel MOSFETs in the positive lead keep B- and P- at nearly the same potential, so the CAN and UART ground stays common with the host. Low-side switching is cheaper but breaks the common ground whenever the pack trips. Decided: high side.

3. **Microcontroller and firmware base.** Option A: STM32G0B1 class with built-in CAN, building on the open [Libre Solar BMS firmware](#) (Apache-2.0, supports bq769x2 front ends). Option B: ESP32-C6 class with Wi-Fi and Bluetooth for a phone app, at higher idle current. Option C: a new firmware from scratch. Decided: A. Reused Apache-2.0 files keep their license and notices in a separate folder, `firmware/third_party/`, and are listed in `LICENSE-SOFTWARE`; CellGuard's own code stays MIT. No firmware exists yet: writing it is TRL 4 work, on hold.
4. **40 A continuous rating.** Covers a 30 A vehicle controller and a 400 W inverter on a 12.8 V pack. A 100 A variant would need more MOSFETs, heavier copper and a larger plate. Decided: 40 A for the first board.
5. **Passive balancing at 100 mA.** Simple and cheap; adequate for packs up to about 100 Ah. Active balancing is out of scope.
6. **Independent secondary protector.** A BQ77216-class protector and SCP fuse close the single-fault gap (R4) for about \$10, at the cost of about 0.6 W in the power path. Decided.
7. **Fuse on the base plate.** Keeps the fuse, switches and sensing together as one tested assembly, at the cost of a longer plate. Kept.
8. **CAN message set.** CellGuard carries the SwapCell interface v0.3 message set (250 kbit/s) as a profile, so SwapCell hosts such as PowerBox and SunSpoke can read a CellGuard pack without changes. Decided, subject to the SwapCell project agreeing (cross-repo action).
9. **Chemistry scope.** Hardware for LFP and for an NMC profile up to 14 cells (under 60 V), so CellGuard could serve as SwapCell's 13S BMS; LFP firmware first. Decided, subject to the SwapCell project (cross-repo action). An NMC build needs the NMC threshold option of the secondary protector and SwapCell's INTERLOCK wake input.

■ Key numbers

These values come from CGD-CAL-001, which states every assumption; they are calculations, not measurements.

Table 2. Key numbers at TRL 3.

Quantity	Value	Requirement
Board loss at 40 A, 1.5 mΩ-class switches (MOSFETs 1.41 W, shunt 0.40 W, copper 1.02 W, SCP fuse 0.64 W)	3.47 W	R6, 4 W: met (was 4.4 W with 2.5 mΩ parts)
External fuse and links at 40 A	1.60 W	
Power path efficiency, 16S at 40 A	99.75 %	
Plate temperature rise at 40 A	10.4 K	
MOSFET junction at 40 A and 40 °C ambient	51.0 °C (TOLT), 55.4 °C (TOLL)	R5, 110 °C: met
MOSFET junction bound, 80 A for 10 s	54.4 °C (TOLT), 77.6 °C (TOLL)	R5: met
Protection settings (decided, CGD-DDR-002)	SCD 400 A in 15 μs, OCD2 160 A in 20 ms, OCD1 96 A in 320 ms, OCC 48 A	R3: met on paper
Prospective short-circuit current	716 A (4S 20 Ah) to 4,976 A (16S 280 Ah)	Fuse breaking capacity 10 kA
Balancing	103 mA, 0.350 W per channel; 1 % on 100 Ah in 9.7 h, on 280 Ah in 27.2 h	R7: at risk (resistor hotspot 73.9 °C)
State-of-charge error after five days between prompted full charges	8.7 points (20 Ah), 6.7 points (100 Ah), with a calibrated coulomb counter offset	R8, 10 points between prompted full charges: met with calibration (24.5 points on 20 Ah uncalibrated)
Current error at 80 A after one-point calibration	0.404 A	R9, 0.850 A: met
Sleep and ship-mode current	54.5 μA and 6.0 μA	R10: met
CAN bus load, 250 kbit/s	1.84 %	R11: met
Log capacity	65,536 records of 32 bytes	R12: met
Precharge of 2 mF to 90 %	0.46 s, 3.38 J, 34.1 W peak	R13: met
Size and mass	220 x 110 x 33 mm; 0.664 kg	R14, 0.7 kg (relaxed 2026-10-02): met
Parts cost	\$159.15 estimated (113.7 % of the \$140 value-engineering target)	R16: over the value-engineering target by \$19.15

State-of-charge method

LFP's flat voltage curve means voltage alone cannot give state of charge in the middle of the range. The concept uses coulomb counting from the front end's integrated coulomb counter, anchored at two points where LFP voltage does change quickly: full (all cells above 3.45 V with charge current tapering below C/20) and near empty (lowest cell below 3.0 V at light load). Between two anchors the firmware learns the usable capacity, which also feeds the state-of-health log. After a rest of two hours or more at the ends of the range, the open-circuit voltage is used as a

check. The first error budget is in CGD-CAL-001 section 6: the coulomb counter's offset sets the drift, so on a 20 Ah pack the offset must stay under about 1.5 μV across the shunt (about 6 mA) to hold 10 points over the five days after which the firmware prompts for a full charge (decided 2026-10-02).

■ Safety

Safety: CellGuard connects directly to lithium cells that can deliver thousands of amperes into a short circuit. Treat every build as live. Use insulated tools, cover exposed terminals and busbars, and fit the pack fuse before connecting any load.

Safety: Connect the balance harness only in the order given by the front-end maker, with the board not yet connected to the power leads, and check every sense lead voltage with a meter first. A wrong or loose sense lead can make the board report a false cell voltage, which can lead to overcharge.

Safety: First power-up and all threshold changes should be tested on a cell simulator or a bench supply with current limiting, not on a full pack. Charge test packs on a non-combustible surface, inside a fire-resistant enclosure where possible, never unattended, with a lithium-rated or Class D extinguisher and sand nearby.

Safety: The MOSFETs cannot always interrupt a hard short from a large pack; the fuse must be sized for the pack's prospective short-circuit current (about 5.0 kA for a 16S 280 Ah pack by CGD-CAL-001; specify a 10 kA breaking capacity) and rated for DC at 80 V or more. Set the short-circuit delay to its 15 μs minimum: at 60 μs the current in a large pack reaches about 2.5 kA before the MOSFETs open. A TVS diode across P+ and P- limits the voltage spike when the MOSFETs open under load.

Safety: The secondary protector blows the SCP fuse if a MOSFET shorts or the front end fails. Until an SCP fuse rated for 40 A at 60 V DC is confirmed (R4 at risk), use only chargers with a fixed, correct end-of-charge voltage for the pack. A blown SCP fuse means a fault: inspect the board before repairing it.

Safety: The precharge resistor would overheat if the load stayed shorted; firmware limits precharge to 1 s and reports a fault.

Maximum pack voltage stays below 60 V DC. CellGuard is a research, educational and prototype design. It is not certified and does not replace a certified BMS where a product must meet a standard.

■ Open questions

- [x] Licensing of reused Apache-2.0 firmware: separate `firmware/third_party/` folder with original notices, listed in `LICENSE-SOFTWARE`. Decided by Amish, 2026-09-25: go with recommendation.
- [x] Value-engineering target: `budget_usd` (a hypothetical control target, not a limit) raised from \$120 to \$140. Decided by Amish, 2026-09-25: go with recommendation.
- [x] Responses to R6, R8 and R14: 1.5 m Ω -class switches in the TOLT package (3.47 W, CGD-CAL-001 v0.6); a firmware prompt for a full charge when about five days pass without one, with R8 restated to apply between prompted full charges; R14 relaxed to 0.7 kg with the 4 mm plate kept. Decided by Amish, 2026-10-02 (CGD-DEC-001).
- [] SwapCell project's agreement to the NMC profile and the message set profile: to be recorded in the SwapCell repo once the board envelope question is settled (SwapCell expects about 230 x 58 x 10 mm; CellGuard is 220 x 110 x 33 mm). Decided by Amish, 2026-10-02 (CGD-DEC-001).

- [x] Cover: printed flame-retardant polycarbonate, not sheet aluminium; opaque for the prototype, with a light pipe over a three-colour status light. Decided by Amish, 2026-10-02 (CGD-DEC-001).
- [x] State-of-charge check: cycle a 4S 20 Ah LFP pack through seven days of partial cycles with a bench supply and a low-cost DC electronic load, count charge with a calibrated reference shunt and meter, then discharge fully to measure the true state of charge. Decided by Amish, 2026-10-02 (CGD-DEC-001).
- [] First test partner: a small off-grid solar installer that builds 16S LFP packs, testing on their bench, is the first candidate to approach. Decided by Amish, 2026-10-02 (CGD-DEC-001).

Concept media: [blueprint sheet](#), [interactive 3D model](#).